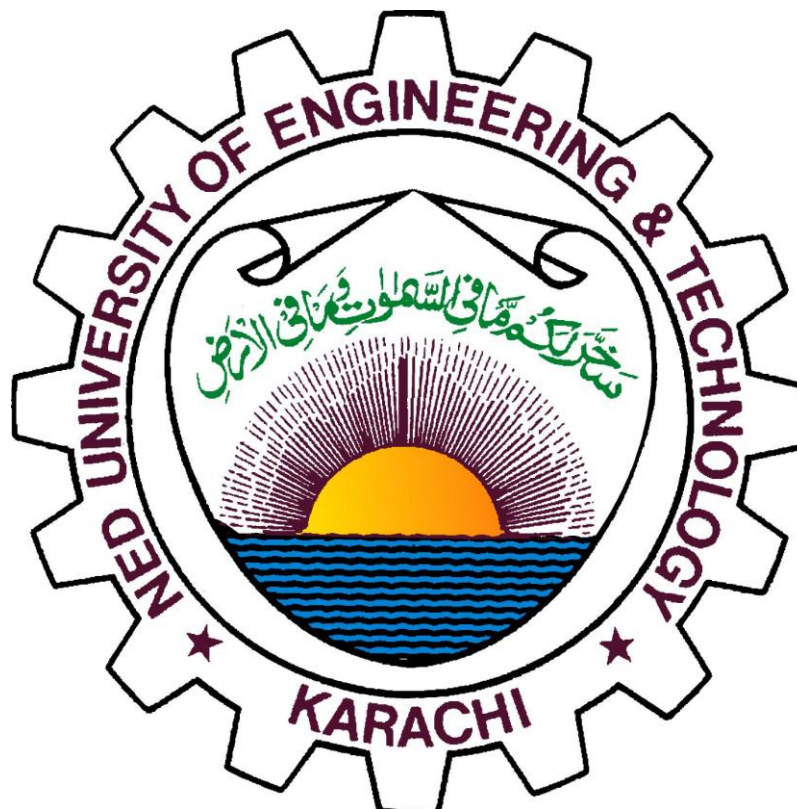


NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Artificial Intelligence And Expert Systems (CT-361)



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CT-24143

Artificial Intelligence – B

----- Lab Assignment 01 -----

SWI PROLOG BASICS

Q.1) A person is healthy if they eat some food and that food is a fruit.

```
healty(Person) :- eat(Person, Food), fruit(Food)
```

Q.2) Two people are siblings if they have the same parent.

```
sibling(X, Y) :- parent(Z, X), parent(Z, Y), X \= Y ▲
```

Q.3) For below facts, rules, and queries write English Meanings.

(1) color(carrots, orange).

The color of carrots is orange.

(2) likes(Person, carrots):-vegetarian(Person).

The person likes carrots, if the person is vegetarian.

(3) pass(Student) :- study_hard(Student).

Student is pass, if the student study hard.

(4) ?- pass(Who).

Who passed ?

(5) ?- teaches(professor, Course).

Professor teaches a course.

(6) enemies(X, Y) :- hates(X, Y), fights(X, Y).

X is an enemy of Y, if X hates Y and X fights Y.

Q.4) Enter the program into Prolog and execute the queries shown below:

Rules and Facts:

```
phone_number(rm303g, 5767).  
phone_number(rm303a, 5949).  
at(dr_jones, rm303g).  
at(dr_mike, rm303a).  
visiting(dr_mike, dr_jones).  
  
ring(Person, Number) :- location(Person, Place), phone_number(Place, Number).  
location(Person, Place) :- at(Person, Place).  
location(Person, Place) :- visiting(Person, Someone), location(Someone, Place).
```

Prolog Queries

```
34 ?- location(dr_bottaci, Pl).  
false.
```

```
8 ?- ring(dr_mike, Number).  
Number = 5949 .
```

```
9 ?- ring(Person, 5767).  
Person = dr_jones .
```

```
10 ?- ring(Person, Number).  
Person = dr_jones,  
Number = 5767 ;  
Person = dr_mike,  
Number = 5949 .
```

```
18 ?- ring(dr_jones, 999).  
false.
```

Q.5) For below english sentence write applicable Prolog facts, rules, and queries.

(1)

```
%Maria reads a logic programming book by author Peter Lucas.  
reads(maria, logical_programming_book).  
author(logical_programming_book, peter_lucas).
```

(2)

```
%Anyone likes shopping if she is a girl.  
girl(Maria).  
girl(Luther).  
likes(X, shopping) :- girl(X).
```

```
43 ?- likes(maria, shopping).  
true ;  
true.
```

(3)

```
%Who likes shopping?  
girl(anna).  
likes(X, shopping). ▲
```

```
44 ?- likes(X, shopping).  
X = maria .
```

(4)

```
%Kirke hates any city if it is big and crowded.  
hates(kirke, X) :- city(X), big(X), crowded(X).
```

Q.6)

Rules and Facts:

```
woman(jia).  
man(john).  
healthy(john).  
healthy(jia).  
wealthy(john).  
travelar(X) :- healthy(X), wealthy(X).  
can_travel(X) :- travelar(X).
```

Prolog Queries

(1) Who can travel.

```
55 ?- can_travel(X).  
X = john.
```

(2) Who is healthy and Wealthy.

```
56 ?- healthy(X).  
X = john.
```

```
57 ?- wealthy(X).  
X = john.
```

(3) Traveler(john).

```
60 ?- travelar(john).  
true.
```

Q.7)

Rules and Facts:

```
vegetarian(jose).  
vegetarian(james).  
  
vegetable(carrot).  
vegetable(egg_plant).  
  
likes(jose, X) :- vegetable(X).  
  
loves(Who, egg_plant) :- vegetarian(Who).▲
```

Prolog Queries

```
72 ?- vegetable(X).  
X = carrot ;  
X = egg_plant.
```

```
66 ?- vegetable(potato).  
false.
```

```
73 ?- vegetarian(X).  
X = jose ;  
X = james.
```

```
74 ?- likes(jose, What).  
What = carrot ;  
What = egg_plant.
```

```
77 ?- likes(Who, egg_plant).  
Who = jose.
```

```
76 ?- loves(Who, egg_plant).  
Who = jose ;  
Who = james.
```

Q.8)

Rules and Facts:

```
male(paul).  
male(joe).  
female(mary).  
female(jane).  
  
father_of(joe, paul).  
father_of(joe, mary).  
mother_of(jane, paul).  
mother_of(jane, mary).
```

Prolog Queries

```
79 ?- father_of(paul, mary).  
false.
```

```
80 ?- father_of(X, mary).  
X = joe.
```

```
81 ?- mother_of(X, paul).  
X = jane.
```

Q.9)

Rules and Facts:

```
man(ahmed).
woman(sara).
father(ahmed, sara).
parent(ahmed, sara).

married(sara, john).
man(john).

brother(ahmed, ali).
brother(ali, ahmed).
married(ali, fatima).
man(ali).

women(fatima).
mother(amina, ahmed).
parent(amina, ahmed).
woman(amina).

mother(X, Y) :- woman(X), parent(X, Y).
father(X, Y) :- man(X), parent(X, Y).
sister(X, Y) :- woman(X), parent(P, X), parent(P, Y), X \= Y.
brother(X, Y) :- man(X), parent(P, X), parent(P, Y), X \= Y.
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).
uncle(X, Y) :- brother(X, Z), parent(Z, Y).
```

Prolog Queries:

(1) Who is Sara's father?

```
83 ?- father(X, sara).
X = ahmed ;
X = ahmed ;
false.
```

(2) Who is Ahmed's brother?

```
85 ?- brother(ahmed, X).
X = ali ;
false.
```

(3) Who are the children of Ahmed?

```
86 ?- parent(ahmed, X).  
X = sara.
```

(4) Who are Grand Parents of Sara?

```
87 ?- grandparent(X, sara).  
X = amina.
```

(5) Who is the Uncle of Sara?

```
92 ?- uncle(X, sara).  
X = ali ;
```

(6) List all the women in family.

```
96 ?- woman(X).  
X = sara ;  
X = fatima ;  
X = amina.
```

(1) List all the men in family.

```
99 ?- man(X).  
X = ahmed ;  
X = john ;  
X = ali.
```

Family Tree

